A-A-50760
30 MAY 86

COMMERCIAL ITEM DESCRIPTION

SIGNAL GENERATOR

The General Services Administration has authorized the use of this commercial item description

Scope. Rack mounted in the AN/APR-38 Test Set used to test/repair/align the APR-38 Intermediate Frequency Processor (IFP) The IFP is used by the system to compute angle of arrival information from the APR-38 receivers This unit's time of IF pulses is crucial to the total system's operations

Salient characteristics

1 Frequency characteristics

a. Frequency range	7.75 - 512 MHz in six bands Band 1 248 MHz - 512 MHz Band 2 124 MHz - 256 MHz Band 3 62 MHz - 128 MHz Band 4 31 MHz - 64 MHz Band 5 15.5 MHz - 32 MHz Band 6 7.75 MHz - 16 MHz
b. Frequency vernier range	> 300 ppm
c. Frequency resolution	Continuously tunable Six digit frequency display with a least significant digit of 100 Hz (7.75 - 62 MHz) or 1kHz (62 - 512 MHz)
d Frequency accuracy	
Locked mode	±(0.05 of last digit time base accuracy)
Unlocked mode	±(last digit + time base accuracy)
e Time base accuracy	
Standard	< ±1 ppm after two hour warm up @ 25°C
Option	< ±0.1 ppm after two hour warm up @ 25°C

f Frequency stability	Lock	Unlocked
Time (after two hr warm up)	<1 ppm/24 hrs.	<10 ppm/10 mins.
Temperature (0°C to 55°C)	<0.33 ppm/°C (Option 0.03 ppm/°C)	<60 ppm/°C
Line voltage (±10% change)	<1 ppm	<10 ppm

FSC 6625

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Load (open to short)	None	None
Level change		
10 dB vernier	None	1 ppm
10 dB steps	None	None
Mode change		
CW to AM	None	None
CW to FM		<150 ppm of carrier frequency

g. Frequency lock hold range ±600 ppm or ±10°C min

2 Spectral purity characteristics

a. Residual FM

Mode	CW		FM Up to Max Pk Dev	
Post Detection Bandwidth	300 Hz - 3 kHz	20 Hz - 15 kHz	300 Hz - 3 kHz	20 Hz - 15 kHz
248 - 512 Mhz	<5 Hz	<15 Hz	<9 Hz	<30 Hz
124 - 256 MHz	<3 Hz	<7 Hz	<5 Hz	<15 Hz
7 75 - 126 MHz	<2 Hz	<5 Hz	<3 Hz	<8 Hz

b Spurious signals

Harmonics	>30 dB below carrier
Sub-harmonics	>45 dB below carrier
Non-harmonics	Not perceptible

c Broadband noise 130 dB below carrier in a 1 Hz bandwidth at maximum power level vernier position over entire frequency range

3 RF Power output characteristics

a. Output power control range 20 dBm (2 2V) to -146 dBm (0 01μV) in 10 dB steps with continuous adjustment between steps Output level indicated on attenuator and meter calibrated in dBm and volts into nominal 50Ω load.

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b. RF output level accuracy·

(1) Output level range:	+20 dBm to -10 dBm	-10 dBm to -50 dBm	-50 dBm to -146 dBm
(2) Total inaccuracy as indicated on output meter and step attenuator (including flatness, calib. inaccuracy, attenuator inaccuracy, meter linearity, detector response, etc.)	Up to 125 MHz ±1.0 dB	Up to 125 MHz ±1.5 dB	Up to 125 MHz ±2.2 dB
	Up to 512 MHz ±1.2 dB	Up to 512 MHz ±1.8 dB	Up to 512 MHz ±2.5 dB

c. Auxiliary accuracy specification

Attenuator accuracy	10.15 dB/10 dB cumulative. Up to ± 1 5 dB max.
RF level flatness	Up to 125 MHz: ±0 25 dB Up to 512 MHz ±0.5 dB
Meter linearity	±2% of full scale.
Output impedance	50 Ω nominal

d. Output VSWR — <1 25 @ ≤ 0 dBm, <1 5 @ 0dBm

e. Output connector — Type N female

4. Modulation characteristics.

a Internal AM

Frequency	1 kHz screw driver adjustable ±2% (rear panel).
Range	Zero to 100% indicated on AM meter calibrated in % modulation.
Accuracy of indicated AM	±8% of reading (20% - 100%) over top 3 dB of vernier range. ±16% of reading (20%- 100%) over remaining 7 dB of vernier range
Incidental FM - Up to 125 MHz: (30% AM)	<150 Hz peak
Up to 512 MHz·	<300 Hz peak
AM distortion· (over top 3 dB of vernier range)	<5% up to 70% AM fc <250 MHz <3% up to 30% AM fc >250 MHz <10% up to 70% AM fc > 250 MHz

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b. External AM:

Modulation frequency response	DC to 100 kHz @ 30% AM
Sensitivity	<2 volts RMS for 95% AM
Accuracy of indicated AM	Same as internal AM (1 kHz modulation frequency)
AM distortion	Same as internal AM (Modulation frequency <5 kHz)
Incidental PM (30% AM)	< 0.15 radians peak up to 125 MHz 0.3 radians peak up to 512 MHz
Incidental FM	Equal to incidental PM times modulation frequency
Input impedance	600Ω nominal
Input connector	BNC Female

c Internal FM

Modulation frequency	1 kHz screw driver adjustable ±2% (rear panel)
Deviation	0-1% peak of carrier frequency indicated on meter with full scale ranges of 30, 100, 300 and 1000 kHz peak. Vernier provides continuous control down to zero deviation
Meter accuracy	±5% of full scale

c External FM

Modulating frequency response	
Unlocked	DC - 200 kHz
Locked	100 Hz - 200 kHz
Deviation	>0.5% peak of carrier frequency indicated on meter with full scale ranges 30, 100, 300 and 1000 kHz peak. Vernier provides continuous control down to zero deviation.
Sensitivity:	<2.7V for maximum deviation. Positive going voltage increases frequency

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Distortion:	< 1% at 0.1% deviation over entire modulation frequency range.
Meter accuracy	±5% of full scale up to 0.1% deviation. (50 Hz to 200 kHz)
Incidental AM	> 50 dB below carrier at ±0.1% deviation
Input impedance.	5000 Ω nominal
Input connector	BNC, female

e External pulse modulation

Repetition rate range	10 Hz - 5 kHz
Pulse level	Actual RMS value of pulse peak level indicated by front panel meter
Pulse level accuracy	CW level accuracy plus sample response for pulse width > 4 µsec <1 dB. Response for pulse width > 10 µsec < 0.1 dB. (PRF > 50 Hz, duty cycle > 0 1%).
Rise and fall time	Down to 10 nsec, varying with frequency and pulse width
On/off ratio	80 dB from 775 MHz to 175 MHz, decreasing to 45 dB at 512 MHz
Sensitivity	< 1 volt turns RF on to level indicated on meter.
Input connector	BNC female
Input impedance	50 Ω nominal
RF power off	Output decreased by on/off ratio when AM PULSE pushbotton is pressed
Delay	No greater than 9 nano seconds from trigger leading edge to output pulse leading edge

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5. Counter characteristics.
(Externally applied signal)

a.	Frequency range:	100 Hz to 10 MHz
b.	Frequency resolution	±1 Hz up to 1 MHz with time base set to 1 sec ± 10 Hz up to 10 MHz with time base set to 0.1 sec or ± 1 Hz with time base set to 1 sec and first digit dropped off
c	Count accuracy	± last digit plus time base accuracy
d	Time base accuracy	
	Standard	±1 ppm after two hours warm up @ 25°C
	Option 7	±0.1 ppm after two hours warm up @ 25°C.
e.	Input waveform	Sinusoid trapezoid, or pulse with 50 nsec min width
f.	Sensitivity	200 mV rms
g	Input impedance	10,000 Ω nominal
h	Input connector	BNC, female

6. Other characteristics

a.	1 kHz output	> 1 volt peak with 1% distortion Z_{out} = 1000 Ω nominal
b.	External counter time base frequency standard input	1.00000 MHz (rear panel BNC connector).
	Input impedance	10 kΩ nominal
	Minimum input signal level	100 mV rms
c.	Dimensions = (Rack Mount, 19")	5 1/4" high (max) X 16 3/4" wide X 22" depth (max) behind front panel
d	Weight	50 pounds (max)
e	Power	115 or 230 volts (± 10%) 47 to 420 Hz, 60 watts (max).

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Contractor certificate. Bid samples shall be tested at San Antonio Air Logistics Center for acceptance testing, marked. "NOT FOR STOCK."

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling and marking. The preservation, packaging, packing, labeling and marking shall be as specified in the contract or order

Known acceptance product(s) The following product(s) are known to meet the requirements of this CID

Manufacturer	Model	NSN
Singer	P/N 6201/1003 Spec 2051 (no longer made)	6625-01-036-8399

UL Standard 1236 is available from Underwriters' Laboratories (UL), Inc 1285 Walt Whitman Road, Melville Long Island New York 11747

NOTE Acceptable bidder's item must be equal to or better than the salient characteristics This includes quality of material, workmanship, etc

Custodians
Air Force - 99
Navy - EC
DLA - ES

Preparing activity
Air Force - 82

(Project 6625-0650)

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